A MORPHISM OF AFFINE SCHEMES IS A CLOSED IMMERSION EXACTLY WHEN IT LIFTS AGAINST EVERY INJECTIVE RING MAP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is the whole lemma environment lines 23495-23507, including the commutative square with the dotted lift; proof span is lines 23509-23544.

Editorial note. The proof contains two xymatrix displays; the retained preamble provides the Xy-pic setup.

Editorial note. The proof cites no label and no bibliography entry, so no external_references are needed.

Editorial note. The original text contains the typographical quirk '\medskip\noindent' between paragraphs and the phrase 'and we win'; both are retained verbatim.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The dictionary between closed immersions of affine schemes and surjective ring maps, pullback squares of spectra, and the Yoneda/anti-equivalence bookkeeping. All are standard; both directions of the criterion are retained.

Lemma 0.1. *A morphism $f : X \to Y$ of affine schemes is a closed immersion if and only if for every injective ring map $A \to B$ and commutative square*

$$\begin{array}{ccc} \mathrm{Spec}(B) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

there exists a lift $\mathrm{Spec}(A) \to X$ *making the two triangles commute.*

Proof. Let the morphism f be given by the ring map $\phi : R \to S$. Then f is a closed immersion if and only if ϕ is surjective.

First, we assume that ϕ is surjective. Let $\psi : A \to B$ be an injective ring map, and suppose we are given a commutative diagram

$$\begin{array}{ccc} R & \xrightarrow{\alpha} & A \\ \downarrow{\scriptstyle \phi} & \nearrow{\scriptstyle \beta} & \downarrow{\scriptstyle \psi} \\ S & \longrightarrow & B \end{array}$$

Then we define a lift $S \to A$ by $s \mapsto \alpha(r)$, where $r \in R$ is such that $\phi(r) = s$. This is well-defined because ψ is injective and the square commutes. Since taking

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the ring spectrum defines an anti-equivalence between commutative rings and affine schemes, the desired lifting property for f holds.

Next, we assume that ϕ has lifts against all injective ring maps $\psi : A \to B$. Note that $\phi(R)$ is a subring of S, so we obtain a commutative square

$$\begin{array}{ccc} R & \longrightarrow & \phi(R) \\ \downarrow{\scriptstyle \phi} & \nearrow & \downarrow \\ S & = & S \end{array}$$

in which a lift $S \to \phi(R)$ exists. Hence, the inclusion $\phi(R) \to S$ must be an isomorphism, which shows that ϕ is surjective, and we win. $\square$